

BRAC UNIVERSITY
Department of Computer Science and Engineering

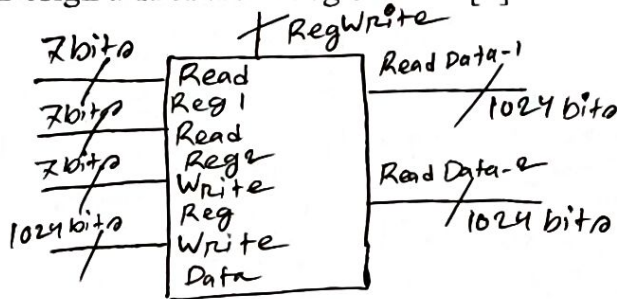
Examination: Quiz - 4
 Duration: 25 minutes

Semester: Spring 2025
 Full Marks: 15

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID: _____	Section: _____
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1. Design a 128X1024 register file. [5]



128 $\rightarrow$ 0 to 127 $\hookrightarrow$ 1111111
 7bits

2. The following table shows the different stages involved in executing instructions and the corresponding durations for each stage:

stages	Instruction Fetch	Register Read	ALU Op	Memory Access	Register Write
Duration	50ps	10ps	30ps	20ps	10ps

Given the above durations, determine the total time required to complete each of the following instructions: [5]

	Instructions	Time to complete each instruction
i.	ADD X_{21}, X_{22}, X_{23}	$50 + 10 + 30 + 10 = 100ps$
ii.	ADDi $X_{21}, X_{22}, 5$	$50 + 10 + 30 + 10 = 100ps$
iii.	LD $X_{21}, 22(X_{20})$	$50 + 10 + 30 + 20 + 10 = 120ps$
iv.	SD $X_5, 22(X_{23})$	$50 + 10 + 30 + 20 = 110ps$
v.	BEQ X_6, X_8, End	$50 + 10 + 30 = 90ps$

Suppose the above instructions are being run in a single-cycle datapath.

a. Determine the clock period of this system. [1]

Answer: 120 ps.

b. Calculate the number of clock cycles needed to execute this instruction set. [2]

Answer: 5

c. What would be the total time to run this instruction sequence? [2]

Answer: $(120 \times 5) = 600 ps.$